



Up to the challenge: Suppression of multiple salient distractors

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Introduction

- The field has focused primarily on studying salient distraction via static singletons (1).
- Historically dynamic stimuli tend to capture attention (2). Whereas static stimuli tend to be suppressed (3).
- Recent evidence suggests that both static and dynamic stimuli can be suppressed but only under specific contexts (4, 5).

How robust is this suppression?

Method

A Search Task

Fixation
until fixation

Search Display
until response

Target
Nonsalient
Distractor
Salient
Distractor

Time

B Search Stimuli

Color

Size

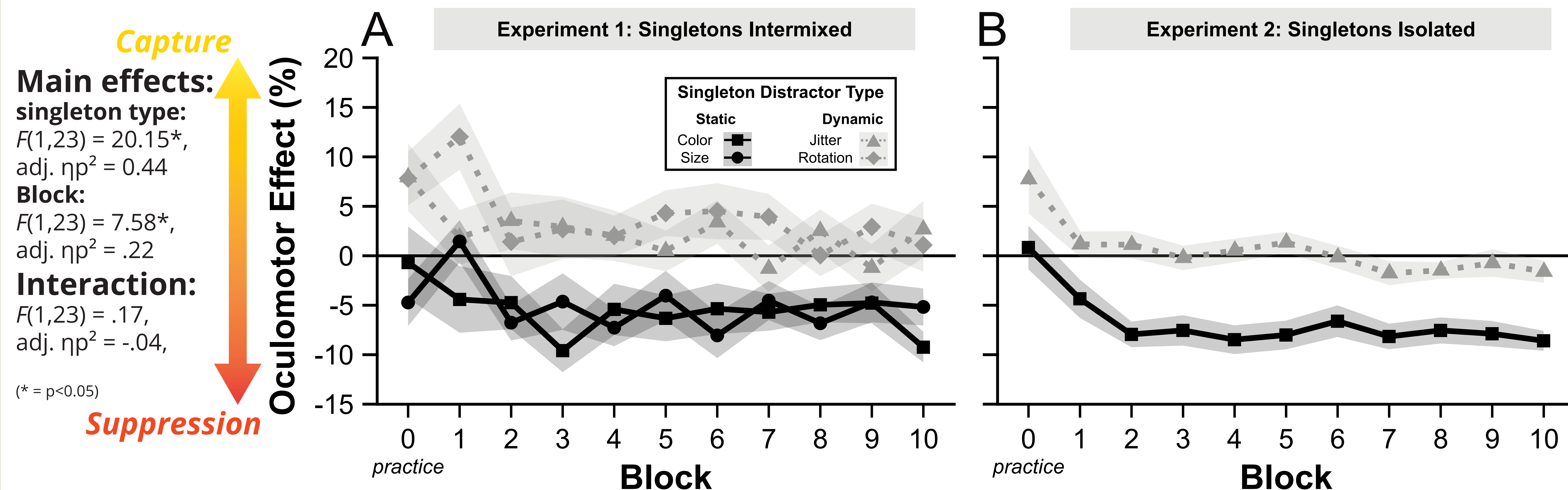
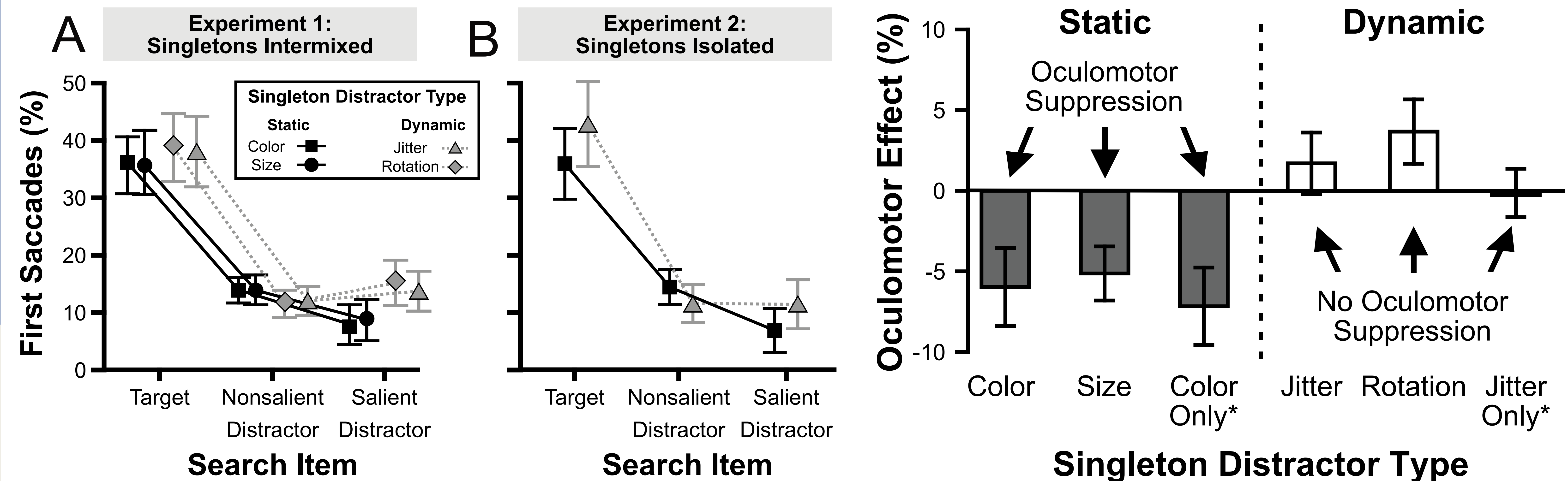
Static

Dynamic

Jitter

Rotation

Results



Main effects:
singleton type:
 $F(1,23) = 20.15^*$,
 $\text{adj. } \eta^2 = 0.44$
Block:
 $F(1,23) = 7.58^*$,
 $\text{adj. } \eta^2 = .22$
Interaction:
 $F(1,23) = .17$,
 $\text{adj. } \eta^2 = -.04$,
(* = $p < 0.05$)

Conclusions

- Only **static** singletons could be reduced **below baseline**.
- However, there was a **reduction in capture** by **dynamic** singletons.
- Intermixing the stimuli led to **“global suppression”** (6).

References

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